

Week 7: Requirement Lifecycle Management

Dr Julian Marx
School of Computing and Information Systems





Announcements

- Assignment 1 is due on Tuesday, 9pm
- Group Member Evaluation via FeedbackFruits on LMS – week 7
- Assignment 2 has been released; walkthrough in Tutorial 6

Assignment 1 Timeline

Task	Suggested Completion
1. Finalise group members and sign up on <i>LMS People - Assignment Group</i>	End of week 3
2. Complete a <i>Governance Plan</i> for your group, which includes allocation of tasks to each member	End of week 3
3. Each student <i>posts</i> the draft work for their section on the LMS Group Homepage - Discussions	End of week 4
4. Each student <i>reviews</i> the draft work of other members and posts their feedback on the LMS Group Homepage	End of week 5
➤ Post your full draft report on the LMS Group Homepage	Tuesday of week 6
➤ Teaching staff reviews full draft reports and provides high-level, general feedback	Wednesday of week 6
5. Prepare and submit the <i>Group Report</i>	Start of week 7
6. Submit the online <i>Group Member Evaluation</i>	End of week 7



Recap: Progress Summary (W1-5)

Frameworks/techniques	Purpose
PEST Analysis	Analyse political (P), economical (E), social (S) and technological (T) influences. <i>(Global/National Contexts)</i>
Porter's Five Forces	Analyse competitive forces affecting the profitability of an organization and industry attractiveness of a business. <i>(Industry Context)</i>
Business Model Canvas	Analyse how an enterprise intends to create value. <i>(Organizational Context)</i>
SWOT Analysis	Evaluate an organization's strengths, weaknesses, opportunities, and threats. <i>(Internal and External Contexts)</i>
POPIT Model	Identify the elements that need to be considered when working with business changes. <i>(Scope)</i>
Stakeholder Onion Diagram	Analyse stakeholders based on their level of involvement. <i>(Stakeholder Analysis)</i>
Stakeholder Matix	Analyse stakeholders based on their level of influence and interest. <i>(Stakeholder Analysis)</i>
5 Whys	Explore the root cause of a particular problem. <i>(Root-Cause Analysis)</i>
Fishbone Diagram	Identify and organize potential causes of a problem or issue. <i>(Root-Cause Analysis)</i>
Pareto Principle (80/20 Rule)	Understand which cause(s) to focus most of your effort on. <i>(Root-Cause Analysis)</i>



Lecture Outline

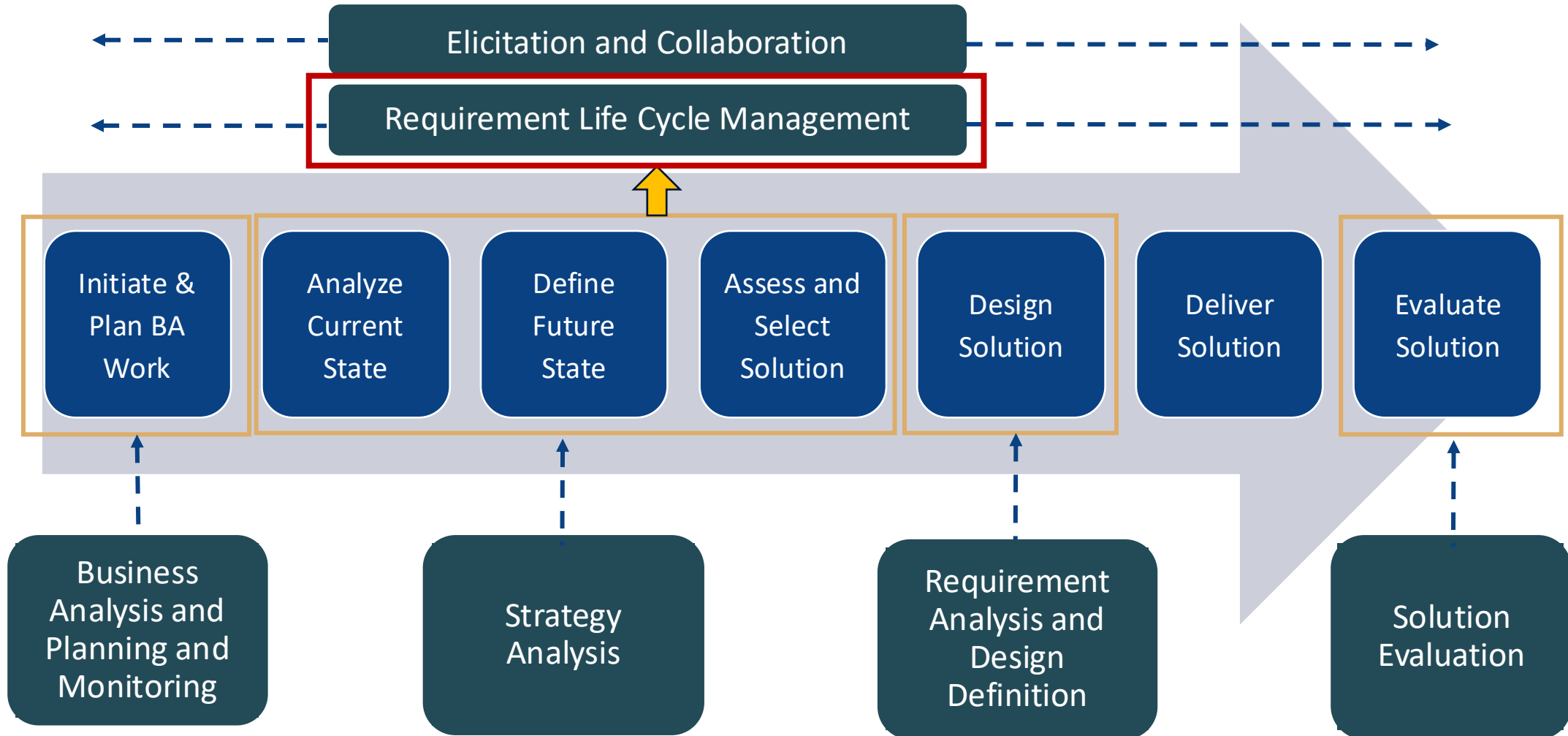
- Requirements – Key Concepts
 - Needs versus Requirements
 - Attributes of a Good Requirement
 - Types of Requirements
- Requirement Lifecycle Management
 - Trace, Maintain, Prioritize, Assess and Approve [TMPAA]
 - Bonus: RLM with Generative and Agentic AI

BABOK (IIBA 2015) Chapters 2&5, Milani (2019) Chapter 15

ISYS90049 Topic Overview



Business Analysis Process & Knowledge Areas



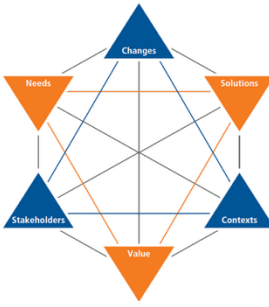


THE UNIVERSITY OF
MELBOURNE

Requirements - Key Concepts

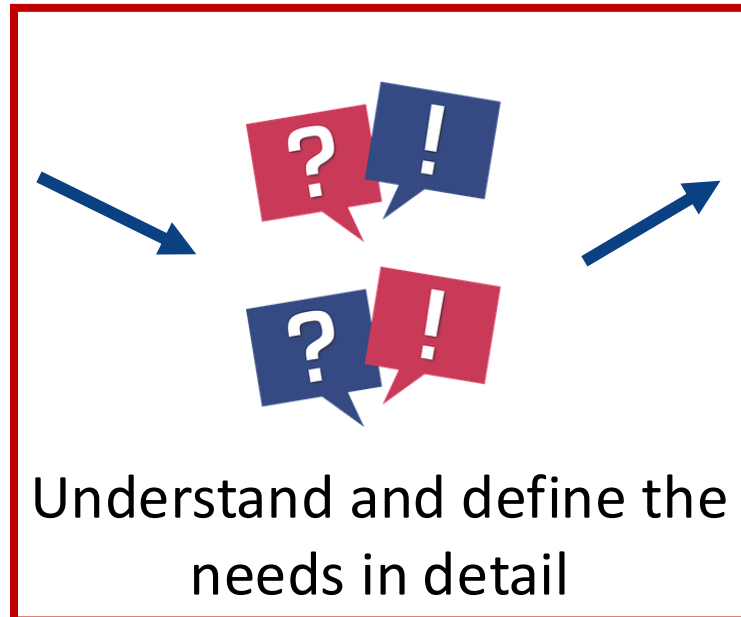


What is Business Analysis? *Revisit*

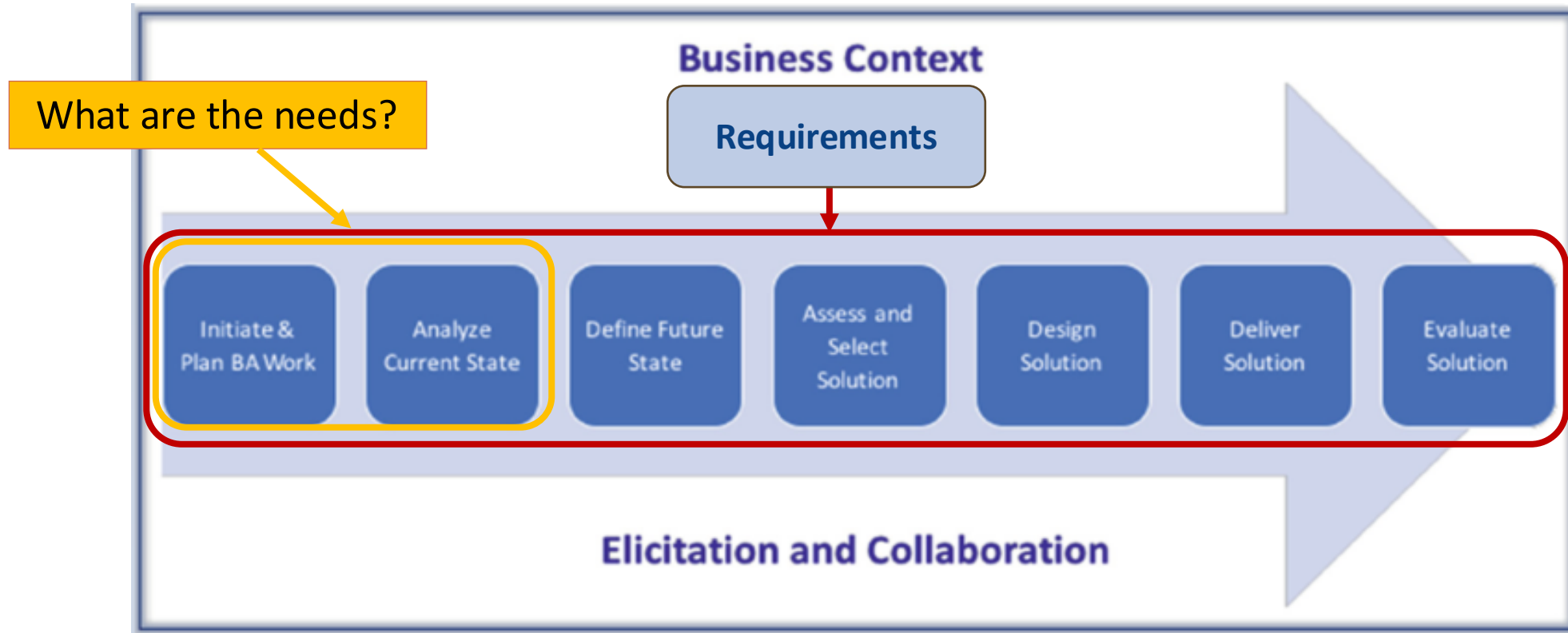


“Business analysis is the practice of enabling **change** in an enterprise by defining **needs** and recommending **solutions** that deliver **value** to **stakeholders**”

Milani (2019), p.3



Business Analysis Process

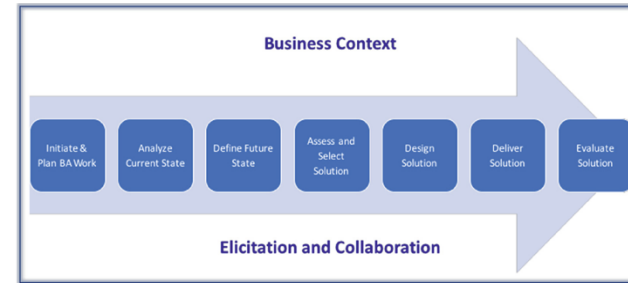
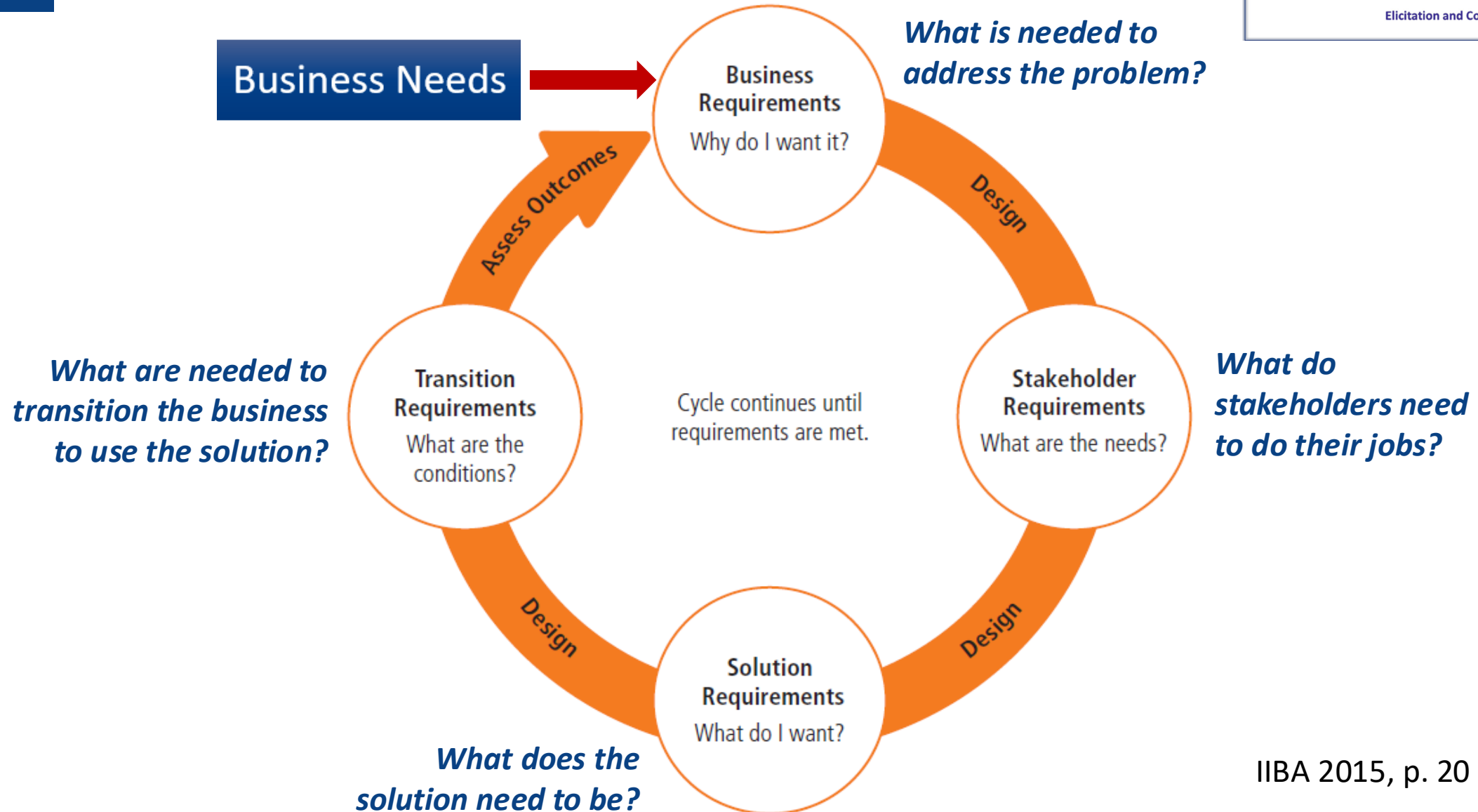


Business Needs vs. Business Requirements

- The line between these two concepts is fuzzy.
 - *Business Need* is related to the “why” questions.
 - *Business Requirement* addresses the “how” questions at a high-level.
- A requirement is a usable representation of a need.
- Requirements focus on understanding what kind of *value* could be delivered if a requirement is fulfilled (IIBA 2015, p. 15)

We need to ... (because of existing problems or opportunities)	Business Requirements
... reduce our IT costs	Merge IT systems
... increase process efficiency	Replace manual activities
... improve customer satisfaction	Improve our customer service and support

Requirements and Design Cycle



Types of Requirements

1. **Business requirements**: statements of **goals, objectives, and outcomes** that describe why a change has been initiated **at the business level**.
E.g., MyGrocer requires better control of perishable inventory to reduce waste and losses from expired products.
2. **Stakeholder requirements**: the **needs of stakeholders** to achieve the business requirements.
E.g., The store manager needs to identify products that are close to expiry or already expired.
3. **Solution requirements**: the **capabilities and qualities of a solution** that meets the stakeholder requirements.
E.g., The system shall allow the store manager to generate a report listing products that are nearing expiry or have expired.
4. **Transition requirements**: define what is needed to transition from **a project** to an **operational solution**.
E.g., Store staff must be trained to record expiry dates correctly in the new system before the expiry tracking function goes live.

Solution Requirements: *Functional vs. Non-Functional*

■ Functional Requirements

- Describe “how” the solution/system will behave
- Describe the **functionality** of the product/service
- Commonly derived from business and/or stakeholder requirements
- Examples:
 - The summary report is to be produced automatically every Monday 9:00am.
 - The summary report is to be produced on-demand

■ Non-Functional Requirements

- Describes **conditions** in which the system must work
- **Qualities** the system must have
 - ❖ usability, performance, reliability, recovery, availability, data privacy, legislative compliance (including sustainability)
- Examples:
 - The system will produce the reports within 3 seconds
 - The system will comply with “customer data protection act”

Attributes of a Good Requirement

- ✓ **Atomic**: self-contained and capable of being understood independently of other requirements or designs.
- ✓ **Complete**: enough to guide further work and at the appropriate level of detail for work to continue.
- ✓ **Consistent**: aligned with the identified needs of the stakeholders and not conflicting with other requirements.
- ✓ **Concise**: contains no extraneous and unnecessary content
- ✓ **Unambiguous**: the requirement must be clearly stated in such a way to make it clear whether a solution does or does not meet the associated need.
- ✓ **Understandable**: represented using common terminology of the audience
- ✓ **Feasible**: reasonable and possible within the agreed-upon risk, schedule, and budget
- ✓ **Testable**: able to verify that the requirement or design has been fulfilled.

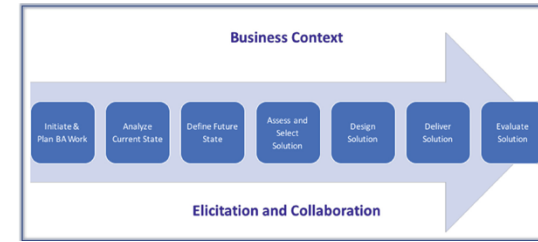


THE UNIVERSITY OF
MELBOURNE

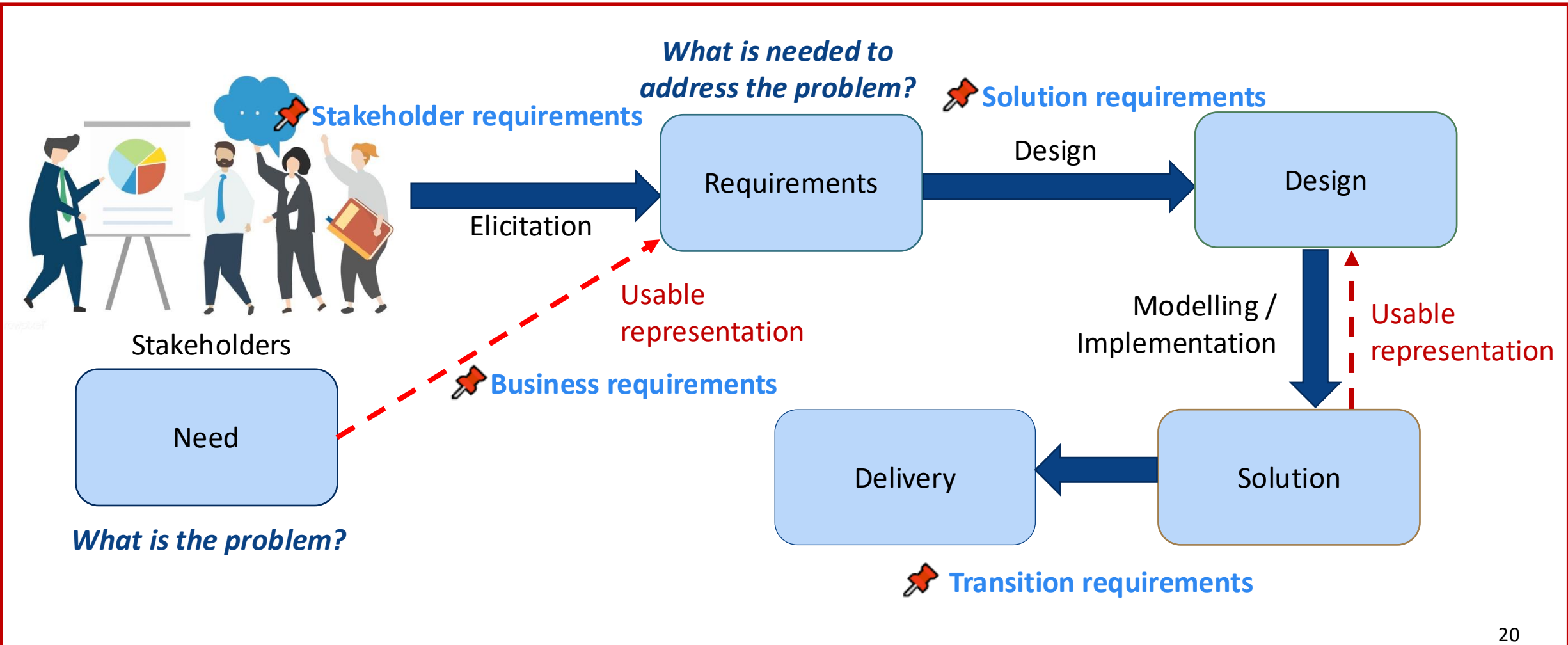
Requirement Lifecycle Management



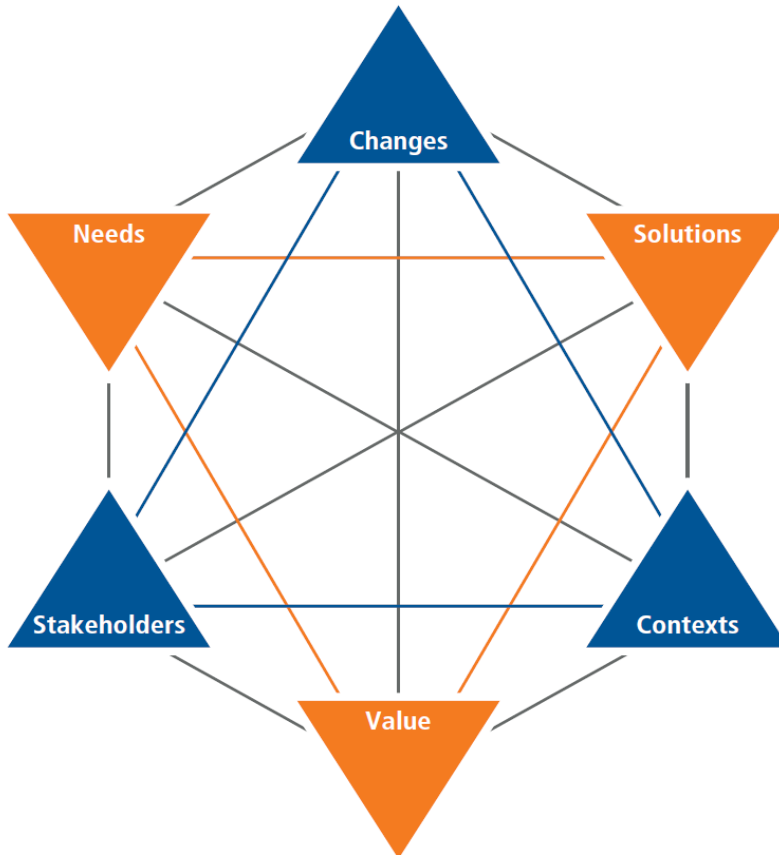
The Big Picture



Requirement Life Cycle Management: *Managing requirements from inception to completion*

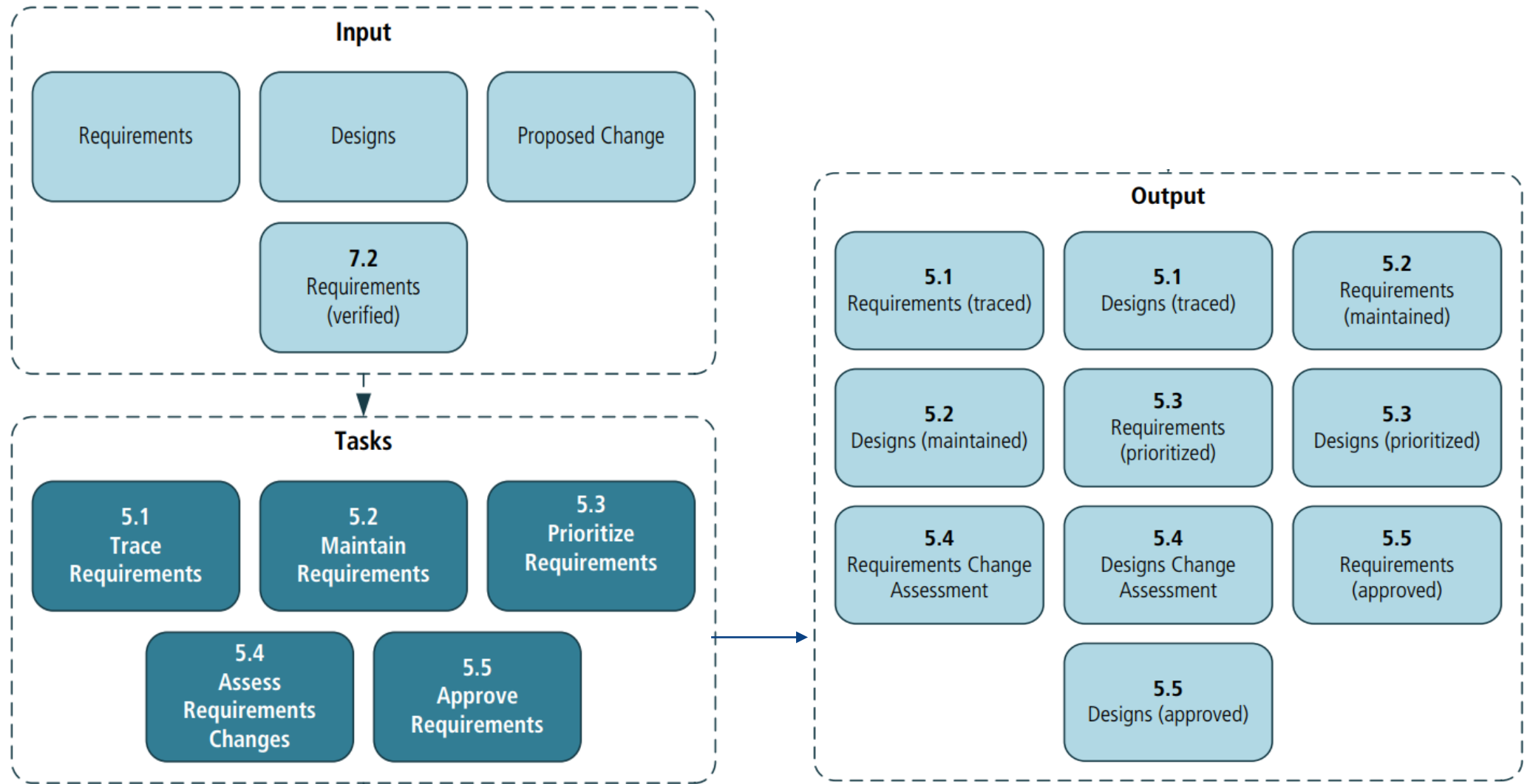


BA Core Concepts in Requirements Life Cycle Management



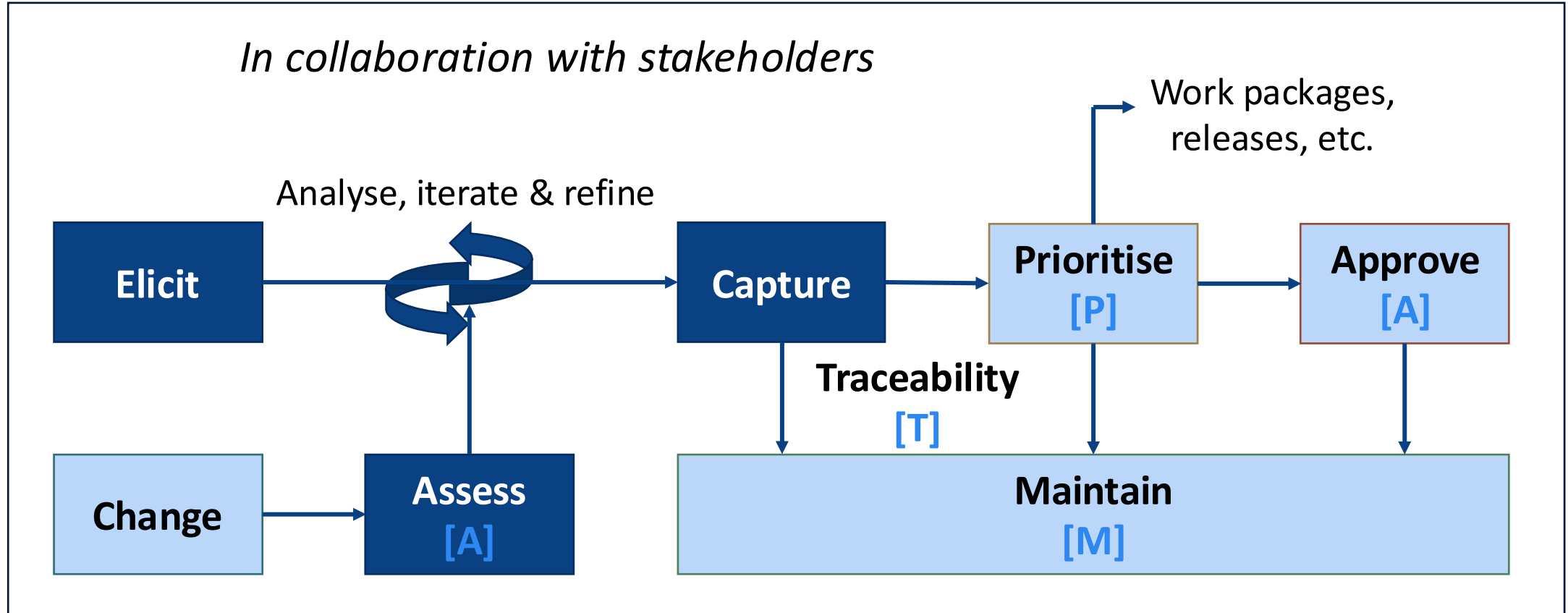
Core Concept	During Requirements Life Cycle Management, business analysts...
Change: the act of transformation in response to a need.	manage how proposed changes to requirements and designs are evaluated during an initiative.
Need: a problem or opportunity to be addressed.	trace, prioritize and maintain requirements to ensure that the need is met.
Solution: a specific way of satisfying one or more needs in a context.	trace requirements and designs to solution components to ensure that the solution satisfies the need.
Stakeholder: a group or individual with a relationship to the change, the need, or the solution.	work closely with key stakeholders to maintain understanding, agreement, and approval of requirements and designs.
Value: the worth, importance, or usefulness of something to a stakeholder within a context.	maintain requirements for reuse to extend value beyond the current initiative.
Context: the circumstances that influence, are influenced by, and provide understanding of the change.	analyze the context to support tracing and prioritization activities.

Requirements Life Cycle Management Input/Output Diagram



[TMPAA]

Managing Requirements (Milani 2019)

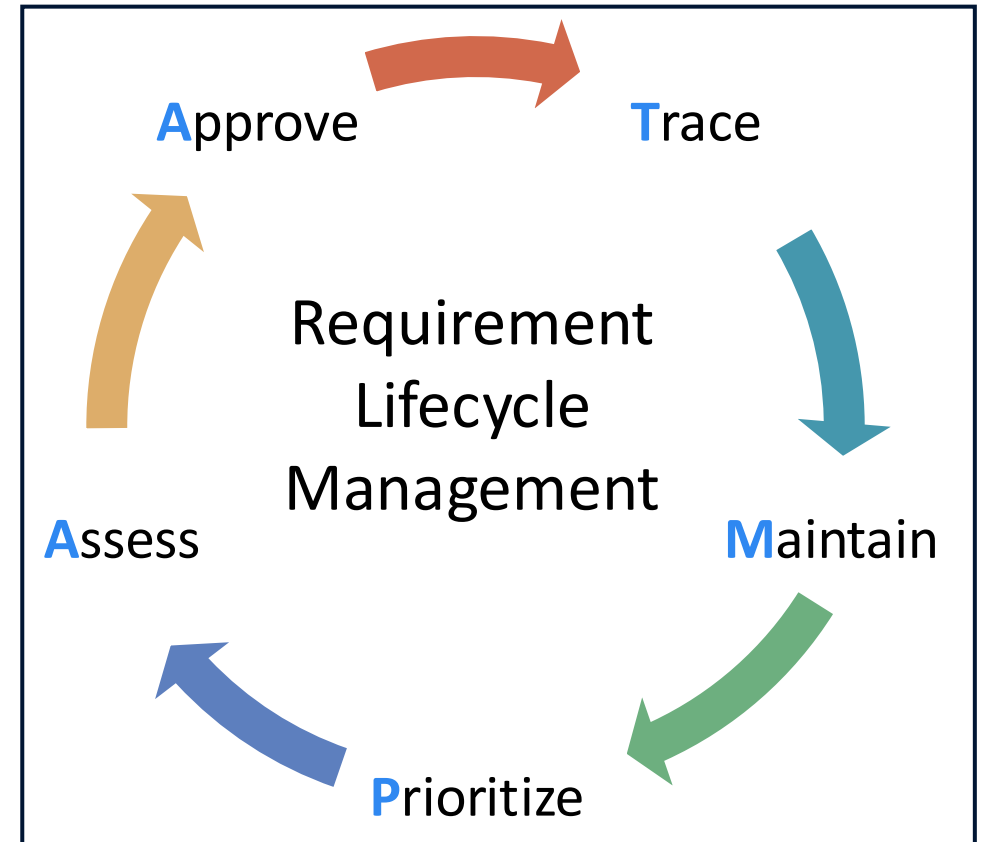


Key tasks: [TMPAA]

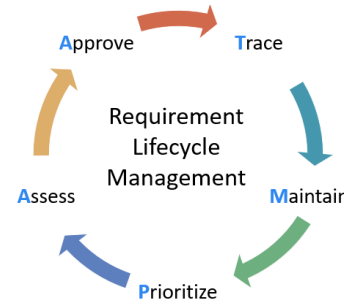
Requirement Life Cycle Management

[Acronym: TMPAA]

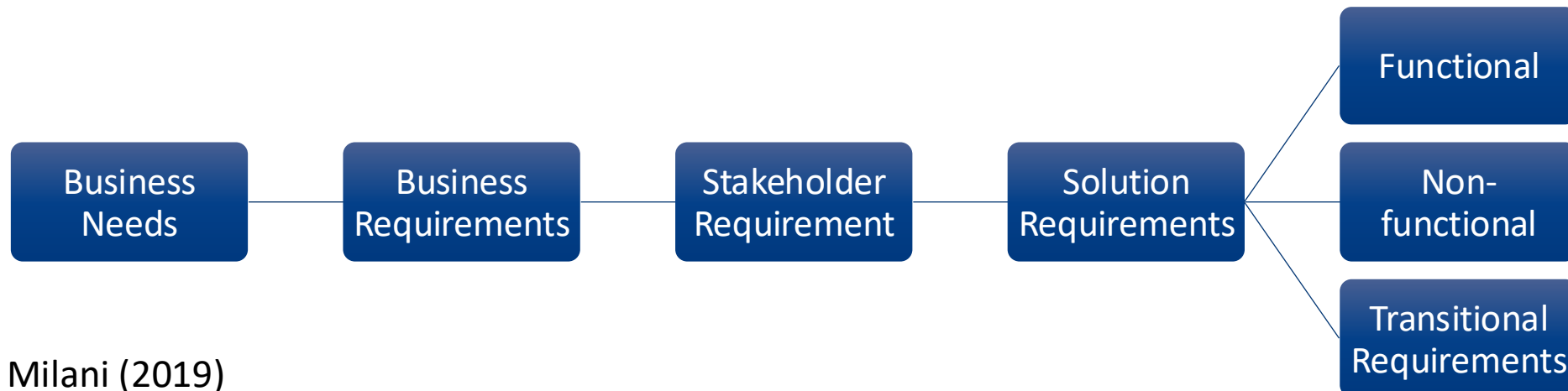
- Describes how requirements and designs are **managed and maintained** from the time they are created until they are retired.
- Establishes **meaningful relationships** between related requirements and designs
- Ensures business, stakeholder, and solution requirements are **aligned** with each other, **prioritized, approved, and maintained** for reuse.



[T] Trace Requirements



- Ensure that **requirements** at *different levels* are **aligned to one another**, and to **manage** the *effects of change* to one level on related requirements (IIBA 2015, p. 79)
- Traceability enables:
 - faster and simpler impact analysis
 - more reliable discovery of inconsistencies and gaps in requirements
 - deeper insights into the scope and complexity of a change
 - reliable assessment of which requirements have been addressed and which have not.



Relationships between Requirements

Derive

- Requirement derived from another

Depend

- Necessity: only meaningful if another requirement is implemented
- Effort: easier to implement if another is also implemented

Validate

- Requirement and another element (a test case) that validates the requirement (determines if it is successful or not)

Tools: Jira

- Jira <https://www.atlassian.com/software/jira>



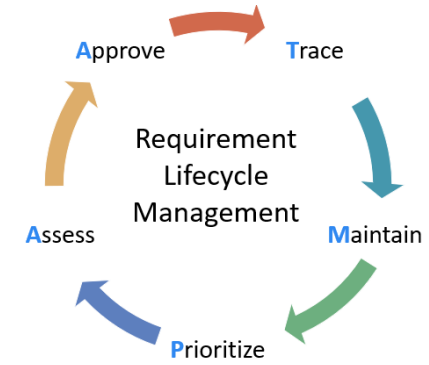
<https://youtu.be/Z-a1RB9HvDI?si=CkFU4dXisdVlv6Lv>



- ## Requirements Traceability Matrix

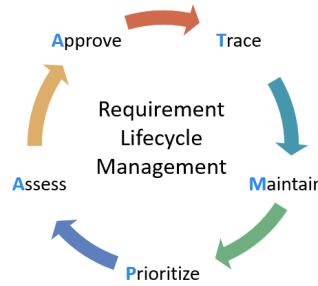
© 2021 International Institute of Business Analysis. IIBA and the IIBA logo are registered trademarks of the International Institute of Business Analysis.

[M] Maintain Requirements



- Ensure that requirements are **accurate** and **consistent**.
- When **new requirements** are considered, the **existing approved requirements** need to be checked to ensure that there is **no conflict**.
- Requirements should:
 - Be represented in the **same format**. If text, the **same wording** should be used. If a model, the **same notation** should be maintained.
 - Follow a **standardized process** for review and approval that ensures **quality**
 - Be easily **accessible** and **understandable**
- Support **reuse** in the current or similar projects

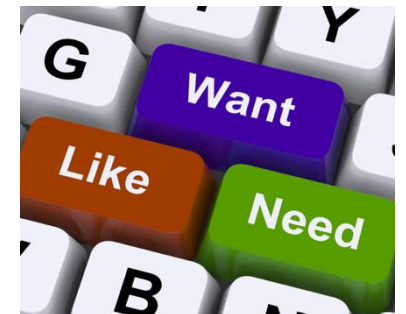
[P] Prioritize Requirements



- Ranking requirements by their **relative importance** to stakeholders – *value* or *sequence*.
- Prioritization is an **ongoing process**, with priorities changing as the context changes (IIBA 2015, p. 87)
- A critical task to ensure the maximum value is achieved.
- Basis for prioritization:
 - **Cost** (effort and resources) versus **Benefit** (advantage)
 - **Risk**: the expected value cannot be delivered
 - **Dependencies**: Some requirements cannot be implemented before others
 - **Penalty**: Negative consequences of not implementing
 - **Regulatory**: Regulatory requirement
 - **Time sensitivity**: Date to be implemented to achieve its maximum value
 - **Stability**: Likelihood of changing over time

Challenges of Prioritization

- Stakeholders commonly need to make **trade-offs** during the prioritization process.
- Examples:
 - *“Everything is high priority. I know that the resources will disappear when this initiative is done, so I need to get these requirements completed while I can.”*
 - *“This functionality is really important to me. I’m sorry if it is not important to you, but I can’t use this solution without it.”*
- **Need vs. want vs. like** - Prioritization should be based on *necessity* of the requirement
 - Is this requirement needed?
 - How often does this occur or is this needed?
 - Apply the 80/20 rule – Pareto rule

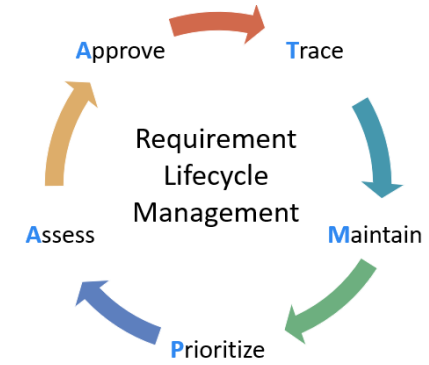


Prioritization Method - Example

Use **MosCoW Method** to mark requirements according to priority

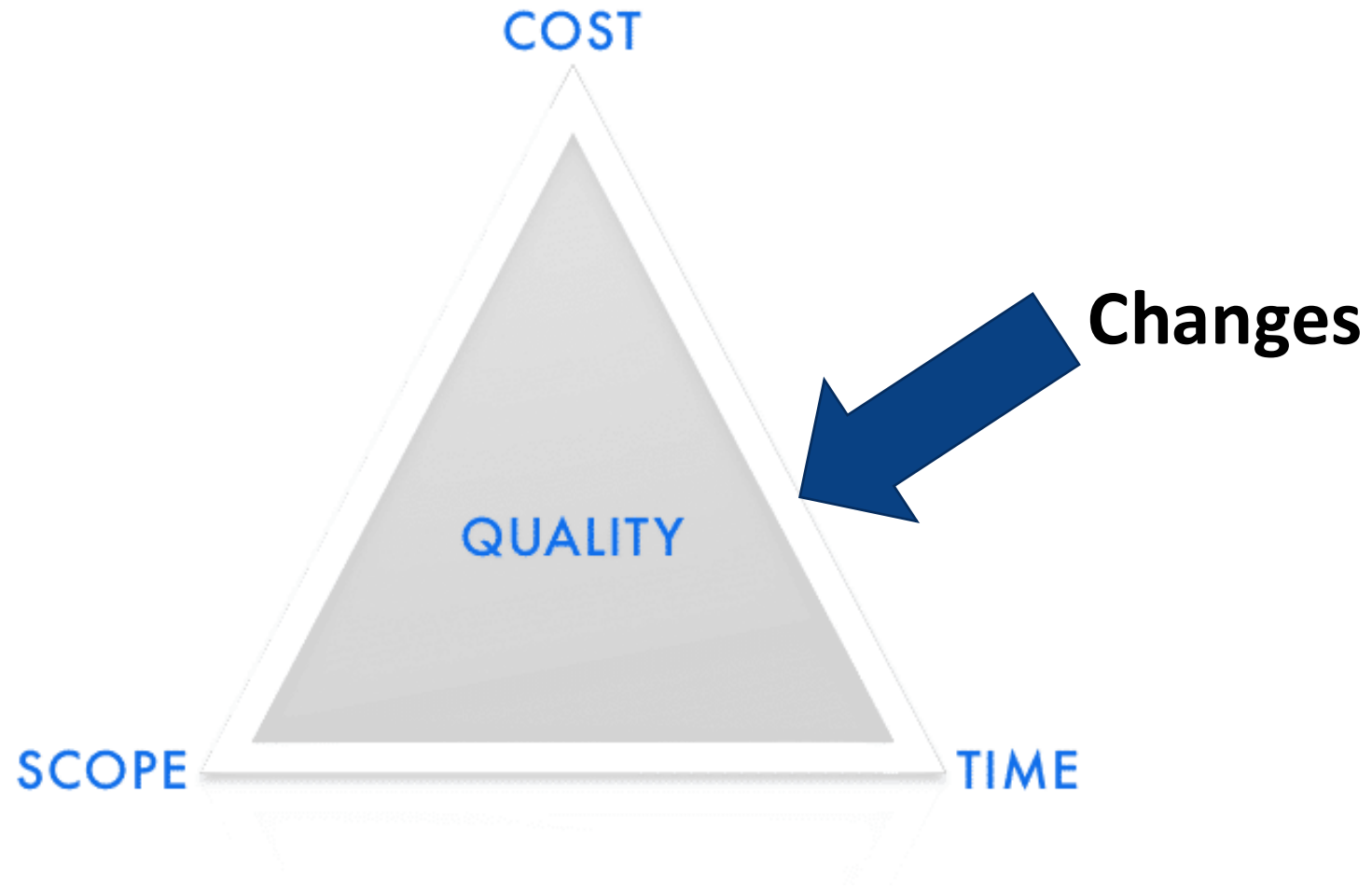
- **M - MUST:** Must have
- **S - SHOULD:** High priority
- **C - COULD:** Desirable but not necessary.
- **W - WON'T:** Will not include now but maybe in the future

[A] Assess Requirements Changes

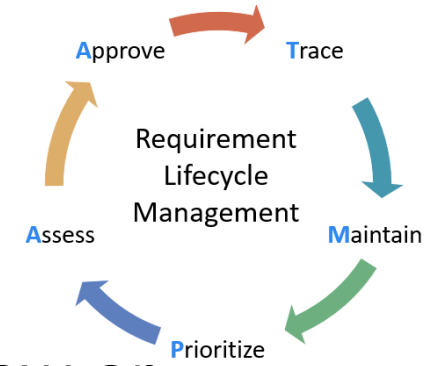


- Evaluate the **impacts** of *proposed changes* to requirements and designs.
- The **formality of assessment** depends on the governance process and the importance of the proposed change.
- **Assessment** of a *proposed change* should consider:
 - Alignment with the overall strategy
 - Impact on value delivered to the business or stakeholder groups
 - Impact on the time to deliver or the resources required to deliver the value
 - Change to risks, opportunities, or constraints associated with the overall initiative
- Document and communicate **all impacts** and resolutions with **all stakeholders**.

Project Management Triangle



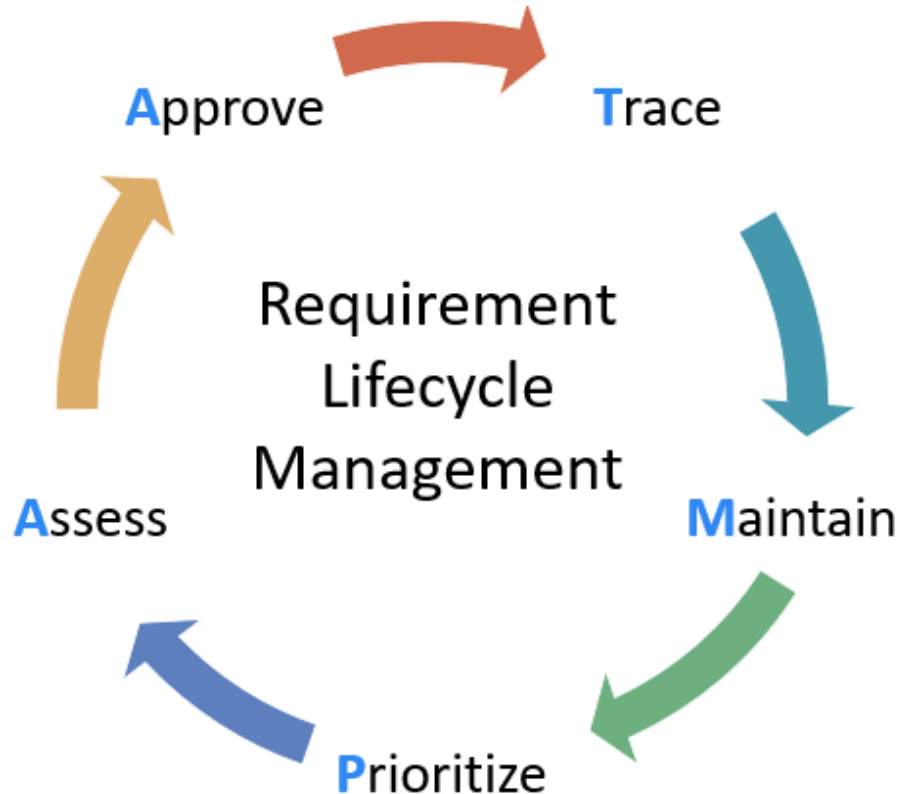
[A] Approve Requirements



- Work with stakeholders to reach **approval** and **agreement** on new or changed requirements. All conflicts and issues must be resolved.
- Approval process is determined by the **Business Analysis Governance**
- Ensure **clear communication** of requirements to the key stakeholders responsible for approving.
- Approval may be formal or informal depending on the approach used (*predictive versus adaptive*).
- Communicate the outcome to all affected stakeholders
- Track and manage the approval

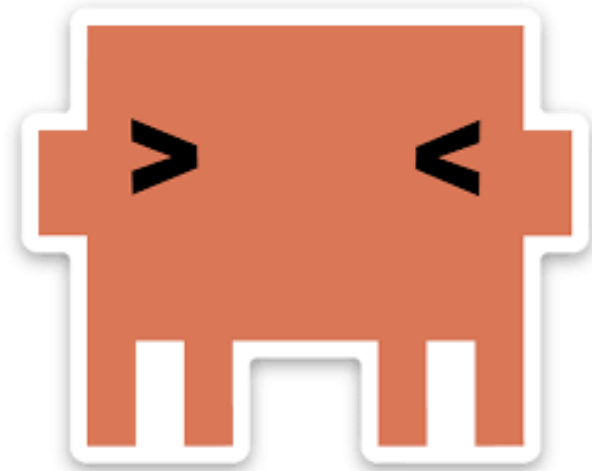
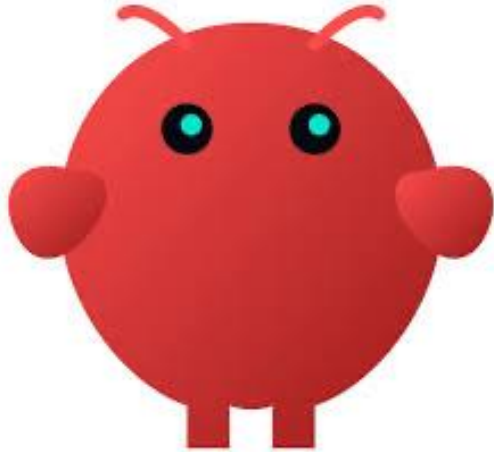
Requirement Life Cycle Management

[TMPAA] Summary



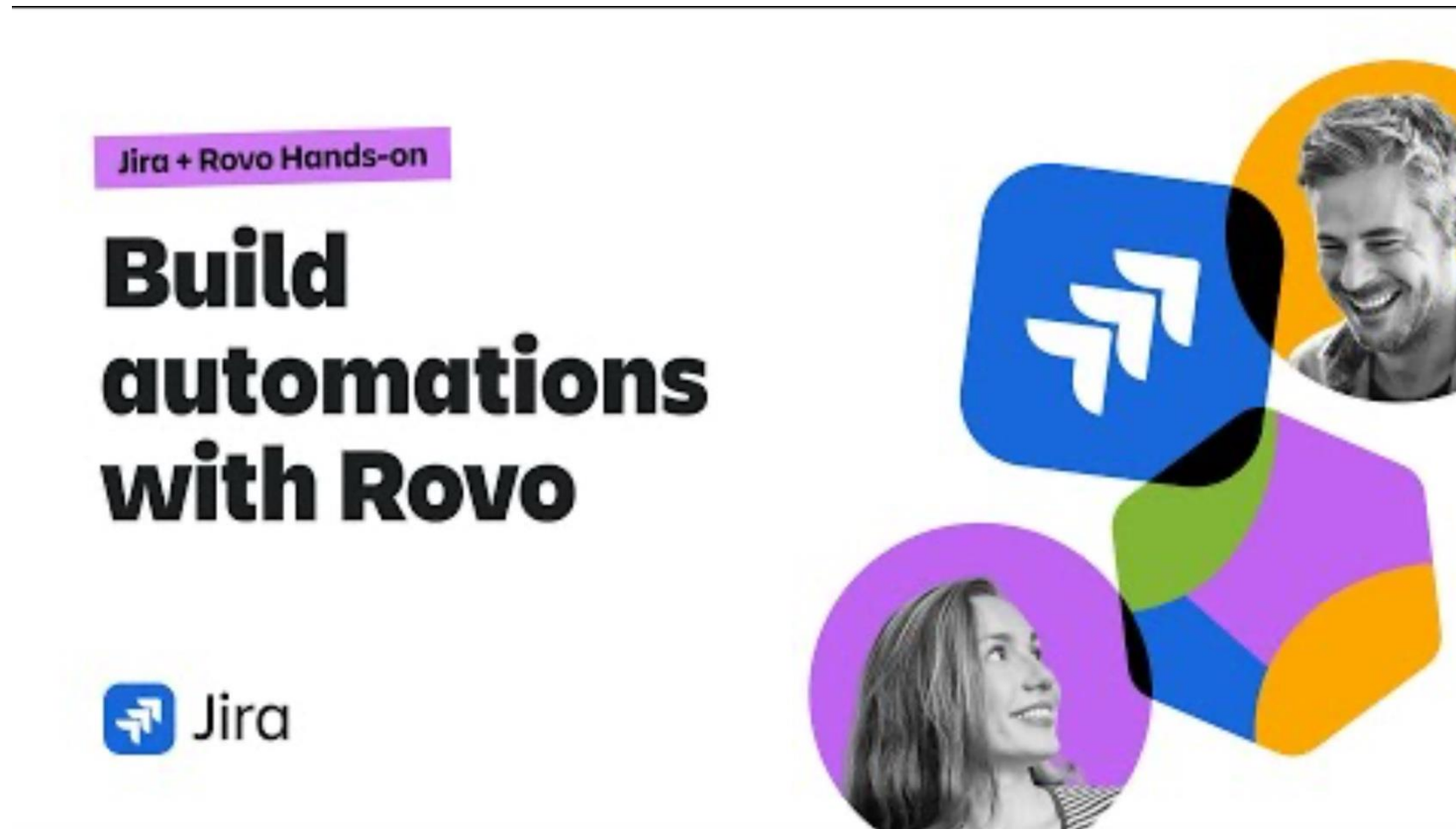
- **Trace Requirements:** Ensure that requirements are **aligned** with each other.
- **Maintain Requirements:** Ensure that requirements are **accurate and consistent**.
- **Prioritize Requirements:** **Rank** requirements by their **relative importance**.
- **Assess Requirement Changes:** Evaluate the **impacts and implications** of **proposed changes** to requirements and designs.
- **Approve Requirements:** Reach **agreement** and **approval** on requirements from key stakeholders.

Bonus: RLM with GenAI and Agentic AI



Bonus: RLM with GenAI and Agentic AI

- Jira + Rovo <https://www.atlassian.com/software/jira>



<https://www.youtube.com/watch?v=FjwGvfl6sNk>

Summary

- **Requirements** are *usable representations of needs* that focus on understanding the potential *value* to be delivered upon their fulfilment.
- There are **different types of requirements** including *business, stakeholder, solution (functional and non-functional)* and *transition* requirements.
- The **Requirements Life Cycle Management** *knowledge area* describes the tasks that business analysts perform to manage and maintain requirements from inception to retirement.
- There are **five tasks** in **requirement life cycle management**: *Trace, Maintain, Prioritize, Assess* and *Approve* [TMPPA].



References

IIBA (2015). *BABOK A Guide to the Business Analysis Body of Knowledge* (3rd ed.). IIBA.

Milani, F. (2019). *Digital Business Analysis*. Springer.